TEK306- Week 37 quiz- solution

Main Objectives

The main objective of this exercise is to be able to describe mobile security countermeasures, technical controls, and security principles.

Description of the Assignment

Attempt each of the below questions as they appear and complete them, preferably as classwork (if you get the time) or later as homework 😊

(1) What is cryptography?

Cryptography is the process or mechanism used for converting ordinary plain text into garbled non-human readable text & vice-versa. It is a means of storing & transmitting information in a specific format so that only those for whom it is planned can understand or process it.

(2) Compare symmetric vs asymmetric encryption.

Symmetric encryption uses the same secret key to encrypt and decrypt information while asymmetric encryption (public-key encryption) uses mathematically linked public- and private-key pairs to encrypt and decrypt senders' and recipients' sensitive data.

(3) What is a hash function?

Hash function is a mathematical algorithm that maps data of arbitrary size to a bit array of a fixed size. It is a one-way function hence practically infeasible to reverse the computation.

(4) Explain digital signature.

Digital signature work by proving that a digital message was not modified from the time it was signed. It is done by generating a unique hash of the message and encrypting it using the sender's private key.

(5) What is cipher-text?

Cryptography helps in securing information in a specific format so that only intended users can understand or process it. When plain text is converted to the unreadable format, that type of text is termed as cipher-text.

(6) Explain secret key?

When plain text is converted to unreadable format through some algorithms, that type of text is termed as cipher text. Cryptographic algorithms are based on mathematical algorithms where these algorithms use the secret key for a secure transformation of data.

(7) What is plain text?

Data which is easily readable & understandable without any special algorithm or method is called plain text (or clear text).

(8) Mention the three types of cryptographic techniques in use.

There are three types of cryptographic techniques used in general. These are Symmetric Key cryptography, public key cryptography, and Hash functions-based cryptography.

(9) What is the name given to conventional cryptography?

Conventional cryptography is also known as secret-key cryptography or symmetric-key encryption.

(10) What is cryptoanalysis?

Cryptanalysis is a study of cracking the cipher-text without knowing the key. This technique is also implemented for designing new cryptographic algorithms or to test their strengths.

(11) Describe the concepts of encryption and decryption.

The process of disguising plaintext in such a way that its substance gets hidden (into what is known as cipher-text) is called encryption. The method of reversing the encrypted text which is known as cipher text to its original form i.e. plain text is known as decryption.

(12) Mention the two categories of mobile security principles.

The two categories of mobile security principles are- principles for mobile device end-users, and principles for mobile application software developers.

Suggested reading:

- The lecture notes.
- https://blog.usejournal.com/cryptography-what-is-it-and-how-does-it-work-2a21a730d694
- https://en.wikipedia.org/wiki/Cryptographic_hash_function